Internal–Leaf Inequalities and a Positive Limiting Leaf Density for Numerical Semigroups with Frobenius Number $2m + 3$

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Abstract

For numerical semigroups of multiplicity m >= 6 and Frobenius number F=2m+3, we study the counts I_m and L_m of internal nodes and leaves in the full numerical-semigroup tree. A seven-configuration model gives the known total N_m=7*2^(m-4), and local sumset conditions yield the inequalities I_m >= 3 L_m for m >= 6 and I_m >= 5 L_m for m >= 12. The factor 3 is attained at m=6; the latter threshold is the least one for which the factor-5 inequality holds for all larger m, although it also holds at m=10. The leaf proportion L_m/N_m tends to a positive constant c with an explicit exponential error bound, and rigorous conditional-Harris estimates give 0.1314883 < c < 0.1446435. These statements are a fixed-family strengthening of Conjecture 41 of Casas, Madrid and Rosales. AI-assisted proof review and independent computations are reported with precise scope; this work has not undergone human peer review and theorem-level novelty is not established.

No.56 - v1.0 preprint. Not human peer-reviewed. Independent AI audit: PASS within its author-reported scope. References: PARTIAL. Novelty: INCONCLUSIVE. Copyright (c) 2026 Ryutaro Yonezu. All rights reserved.

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Introduction

The full tree of numerical semigroups has a child for each minimal generator above the Frobenius number that can be deleted. Casas, Madrid and Rosales [1, Conjecture 41] conjecture that, for fixed admissible multiplicity and Frobenius number, the number of internal semigroups is no smaller than the number of leaves. Here we focus on the infinite subfamily

$$F = 2m + 3, \qquad m \geq 6.$$

Write N_m, I_m, and L_m for the total number of such semigroups and the numbers of internal semigroups and leaves *in the full tree*. Thus $N_m = I_m + L_m$. We establish two strengthened comparisons and a limiting positive leaf proportion, using the same elementary membership-bit model. The total-count formula is **known** and is not advanced as a novelty claim. The `leaf-discriminating trees` of [2] are specially pruned enumeration trees, and their leaves must not be identified with the leaves in the full tree used here.

Seven configurations and the exact leaf criterion

Let S be a numerical semigroup with multiplicity m and Frobenius number $2m + 3$. All positive integers below m and the integer $m + 3$ are absent, while $m, 2m$ and all integers above F are present. Let

$$a_i = \mathbf{1}_{\{m+i\in S\}} \quad (i = 1, 2), \qquad b_i = \mathbf{1}_{\{2m+i\in S\}} \quad (i = 1, 2),$$

and $X_i = \mathbf{1}_{\{m+i\in S\}}$ for $4 \leq i \leq m - 1$. The closure conditions on $(a_1, a_2; b_1, b_2)$ are exactly:

- $(a_1, a_2) = (0, 0)$: any (b_1, b_2) (four possibilities);
- $(a_1, a_2) = (1, 0)$: $(b_1, b_2) = (1, 1)$ (one possibility);

- $(a_1, a_2) = (0, 1)$: $b_2 = 1$, with b_1 free (two possibilities).

The remaining $m - 4$ bits are unrestricted. Thus

$$\boxed{N_m = 7\,2^{m-4}.} \tag{1}$$

This elementary specialization is compatible with the earlier multiparameter counts [2].

A positive element $x > F$ is a minimal generator precisely when it cannot be written as a sum of two positive elements of S. All such candidates belong to $[F+1, F+m] = [2m+4, 3m+3]$. Consequently, S is a leaf precisely when all m integers in that interval have such a decomposition.

Prefix probabilities and finite inequalities

Give the seven initial configurations equal weight. After marginalizing out b_1, b_2, the probabilities of $(X_1, X_2) = (0,0), (1,0), (0,1)$ are $4/7, 1/7, 2/7$, respectively; set $X_3 = 0$. Independently select $X_4, X_5, \ldots$ as fair bits. Define the increasing events

$$E_k = \{X_k = 1\} \cup \{\exists\ i, j \geq 1 :\ i + j = k,\ X_i = X_j = 1\}, \qquad k \geq 4,$$

$$Q_r = \Pr\left(\bigcap_{k=4}^{r} E_k\right). \tag{2}$$

For $4 \leq k \leq m-1$, decomposability of $2m + k$ is equivalent to E_k. The condition for $3m$ is automatic, while the three terminal integers $3m + 1, 3m + 2, 3m + 3$ require separate checks. Thus, writing $p_m = L_m/N_m$,

$$p_m \leq Q_{m-1} \qquad (m \geq 7), \qquad Q_{r+1} \leq Q_r. \tag{3}$$

Theorem A (finite internal–leaf bounds). For the full numerical-semigroup tree,

$$\boxed{I_m \geq 3L_m \quad (m \geq 6),} \qquad \boxed{I_m \geq 5L_m \quad (m \geq 12).} \tag{4}$$

Proof. The exact small certificates in the reproducibility code give

$$Q_6 = \frac{13}{56} < \frac{1}{4}, \qquad Q_{12} = \frac{596}{3584} = \frac{149}{896} < \frac{1}{6}.$$

For $m \geq 7$, (3) gives $p_m < 1/4$, implying $I_m \geq 3L_m$. At $m = 6$, $(I_6, L_6) = (21, 7)$. For $m \geq 13$, (3) gives $p_m < 1/6$, implying $I_m \geq 5L_m$. Direct finite counting establishes the $m = 12$ case, $(I_{12}, L_{12}) = (1508, 284)$. The threshold 12 is minimal in the *eventual* sense because $(I_{11}, L_{11}) = (744, 152)$ violates $I \geq 5L$. The inequality does hold at $m = 10$; therefore 12 is not asserted to be the first individual successful value. The factor 3 is attained at $m = 6$. □

Positive limiting leaf density

For $k \geq 9$, select $t(k) = \lfloor (k-1)/2 \rfloor - 3$ disjoint pairs $(i, k-i)$ with $4 \leq i < k - i$. They involve independent fair bits; X_k is independent of all selected pairs. Failure of E_k requires $X_k = 0$ and every selected pair to fail. Therefore

$$\Pr(E_k^c) \leq u_k := \frac{1}{2}\left(\frac{3}{4}\right)^{\lfloor (k-1)/2 \rfloor - 3}. \tag{5}$$

The series $\sum_{k\geq 9} u_k$ converges. The decreasing sequence (Q_r) therefore has a limit $c \geq 0$. The three terminal decomposability checks can be bounded using disjoint index pairs within $4, \ldots, m-1$, yielding

$$0 \leq Q_{m-1} - p_m \leq 3\left(\frac{3}{4}\right)^{\lfloor m/2 \rfloor - 3} \qquad (m \geq 9). \tag{6}$$

Summing (5) over the infinite tail gives

$$0 \leq Q_{m-1} - c \leq 4\left(\frac{3}{4}\right)^{\lfloor (m-1)/2 \rfloor - 3}. \tag{7}$$

Because both p_m and c lie below Q_{m-1}, the larger of the two deviations in (6)–(7), **not their sum**, bounds $|p_m - c|$. Therefore

$$\boxed{\left|\frac{L_m}{N_m} - c\right| \leq 4\left(\frac{3}{4}\right)^{\lfloor (m-1)/2 \rfloor - 3} \qquad (m \geq 9).} \tag{8}$$

This explicit bound is intentionally conservative: at moderate m it can be numerically weak. It establishes exponential convergence but does not by itself imply that the values at $m \approx 20$ closely approximate c.

It remains to show $c > 0$. **Harris's inequality is applied only after conditioning on the initial (X_1, X_2) pair.** Within each such conditional distribution, the remaining bits are independent and the E_k are increasing events. One can multiply the positive finite-prefix probability by product lower bounds for the subsequent increasing events. Since $\sum u_k$ converges, those products remain positive after a sufficiently long prefix. Averaging over the three initial pairs then gives $c > 0$. Applying Harris directly to the unconditional mixture would not be justified.

Theorem B (limit and quantitative estimate). The limiting leaf proportion $c = \lim_{m\to\infty} p_m$ exists, is strictly positive, and satisfies (8).

Certified numerical enclosure

A separate incremental count yields

$$Q_{32} = \frac{543583947}{3758096384}.$$

Using the conditional Harris inequality for $33 \leq k \leq 70$ and the union bound on the remaining tail,

$$Q_{32} \prod_{k=33}^{70} (1 - u_k)\left(1 - 4\left(\frac{3}{4}\right)^{32}\right) \leq c \leq Q_{32}. \tag{9}$$

Exact rational arithmetic verifies the strict decimal enclosure

$$\boxed{0.1314883 < c < 0.1446435.} \tag{10}$$

The supplied interval verifier checks the arithmetic in (9) conditional on the stated Q_{32}. The C++ verifier separately computes Q_{32}; a user-reported independent AI auditor additionally reproduced that count with a separately written C++ implementation.

Reproducibility and verification scope

The original three source verifiers are preserved byte-for-byte in `repro/`. The Python explicit-membership implementation checks (N_m, L_m) for $m = 6, \ldots, 15$ and finite prefix counts; the C++17 sumset implementation checks $m = 6, \ldots, 24$ and computes Q_{32}. A separate independent AI audit, reported by the author, used Linux, wrote an unrestricted enumeration of numerical semigroups for $m = 6, \ldots, 22$, independently checked prefix counts, and reviewed the conditional Harris reasoning; its verdict within scope was PASS. This was **AI review, not human peer review**.

On 5 November 2026, the author supplied a Windows PowerShell 5.1 transcript reporting successful Python and compiled C++ runs on an unchanged original v0.1 ZIP. The full log is retained in the private audit archive. Its SHA256, verified against the author's stated SHA256, is:

`b3d94afab5686bedeb23b1f961b5b69830faa008284127ff776e2677d0b49fee`

The raw log is deliberately not included in this public reproducibility package because it contains host and local-user identifiers. A matching SHA256 verifies the submitted file's bytes against the reported hash; it does not independently authenticate the entire machine execution process.

Limits and originality status

The result is presented as a *fixed-family strengthening* of Conjecture 41 in [1], not a solution to that conjecture for arbitrary (m, F). The counting formula (1), Harris's inequality, and related sumset probability techniques are not claimed new. Our full-tree leaves are not the specially defined leaves of the leaf-discriminating enumeration trees of [2]. The scope of [3] is the genus-indexed tree, and [4] concerns shapes of random semigroups selected by genus or Frobenius number; neither comparison establishes a comprehensive novelty clearance. **REFERENCES PARTIAL; NOVELTY INCONCLUSIVE.** No human peer review has occurred. All rights reserved.

References

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